Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2022 to 12/31/2022.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Vierra Dairy Farms #2

Physical address of dairy:

Number and Street	City	County	Zip Code
22682 W Williams AVE	Hilmar	Merced	95324

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 01/01/1970

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X045-X190-X032-XXXX

B. OPERATORS

Vierra Dairy LLC

Operator name: Vierra Dairy LLC

Telephone no.: (209) 668-0343 (Landline) (Cellular)

Mailing Address Number and Street	City	State	Zip Code
P.O. Box 1350	Hilmar	CA	95324

This operator is responsible for paying permit fees.

C. OWNERS

R & R Ranches LLC

Legal owner name: R & R Ranches LLC

Telephone no.: (209) 668-0343 (Landline) (Cellular)

Mailing Address Number and Street	City	State	Zip Code
P.O. Box 1131	Turlock	CA	95381

AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	0	798	0	0
Number under roof	474	0	0	0	0	0
Maximum number	480	0	0	798	0	0
Average number	474	0	0	798	0	0
Avg live weight (lbs)	1,400	0	0	700		

Predominant milk cow breed: Holstein

Average milk production: 65 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 19,434.48 *tons per reporting period*

Total nitrogen from manure: 231,063.70 *lbs per reporting period*

After ammonia losses (30% loss applied): 161,744.59 *lbs per reporting period*

Total phosphorus from manure: 39,059.89 *lbs per reporting period*

Total potassium from manure: 89,218.28 *lbs per reporting period*

Total salt from manure: 223,182.90 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 4,950,000 *gallons*

Total nitrogen generated: 24,625.43 *lbs*

Total phosphorus generated: 2,707.23 *lbs*

Total potassium generated: 29,853.19 *lbs*

Total salt generated: 191,145.23 *lbs*

	4,950,000	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	4,950,000	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
T.I.D Canal	Surface water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

No commercial or other nutrient imports entered.

G. NUTRIENT EXPORTS

No solid nutrient exports entered.

No liquid nutrient exports entered.

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
F1	15	15	3	process wastewater	XX45-X190-X032-XXXX
F2	26	26	3	process wastewater	XX45-X190-X030-XXXX
Totals for areas that were used for application	41	41	6		
Totals for areas that were not used for application					
Land application area totals	41	41	6		

B. CROPS AND HARVESTS

F1

Field name: F1

10/26/2021: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 15 Plant date: 10/26/2021

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
03/24/2022	122.70 *ton*	As-is		63.8	9,300.00	1,600.00	23,900.00		13.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	8.18	152.15	26.18	391.00	799.51

04/10/2022: Corn, silage

Crop: Corn, silage Acres planted: 15 Plant date: 04/10/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
08/08/2022	491.55 *ton*	As-is		67.9	4,100.00	1,000.00	5,400.00		5.16

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	32.77	268.71	65.54	353.92	1,085.58

F1

08/16/2022: Sudangrass, silage

Crop: Sudangrass, silage Acres planted: 15 Plant date: 08/16/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/27/2022	118.20 *ton*	As-is		69.8	7,400.00	1,700.00	14,100.00		16.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	8.80	13.60	96.00	0.00
Total actual harvest content	7.88	116.62	26.79	222.22	804.36

F2

Field name: F2

10/26/2021: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 26 Plant date: 10/26/2021

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
03/24/2022	141.44 *ton*	As-is		63.8	9,300.00	1,600.00	23,900.00		13.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	5.44	101.18	17.41	260.03	531.71

04/10/2022: Corn, silage

Crop: Corn, silage Acres planted: 26 Plant date: 04/10/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
08/08/2022	758.42 *ton*	As-is		67.9	4,100.00	1,000.00	5,400.00		5.16

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	29.17	239.19	58.34	315.04	966.32

F2

08/16/2022: Sudangrass, silage

Crop: Sudangrass, silage Acres planted: 26 Plant date: 08/16/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/27/2022	204.88 *ton*	As-is		60.7	9,700.00	900.00	16,600.00		12.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	8.80	13.60	96.00	0.00
Total actual harvest content	7.88	152.87	14.18	261.62	755.63

NUTRIENT BUDGET

A. LAND APPLICATIONS

F1 - 10/26/2021: Oats, silage-soft dough

Field name: F1

Crop: Oats, silage-soft dough — Plant date: 10/26/2021

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/14/2021	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	90.73	3.63	51.73	200.68	108,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	263.82	2,431,815.00 *gal*
Application event totals		90.73	3.63	51.73	464.50	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
12/08/2021	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	30.24	1.21	17.24	66.89	36,000.00 *gal*
Application event totals		30.24	1.21	17.24	66.89	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/19/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	17.62	2.26	26.32	200.28	72,000.00 *gal*
Application event totals		17.62	2.26	26.32	200.28	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/28/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	35.25	4.52	52.63	400.56	144,000.00 *gal*
Application event totals		35.25	4.52	52.63	400.56	

F1 - 04/10/2022: Corn, silage

Field name: F1

Crop: Corn, silage — Plant date: 04/10/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

F1 - 04/10/2022: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/31/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	52.87	6.78	78.95	600.84	216,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	250.32	2,307,372.00 *gal*
Application event totals		52.87	6.78	78.95	851.16	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/15/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	191.63	1,766,378.00 *gal*
Application event totals		0.00	0.00	0.00	191.63	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/26/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	54.08	6.90	72.80	466.65	180,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	141.42	1,303,600.00 *gal*
Application event totals		54.08	6.90	72.80	608.07	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/05/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	252.08	2,323,667.00 *gal*
Application event totals		0.00	0.00	0.00	252.08	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/15/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	43.26	5.52	58.24	373.32	144,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	197.28	1,818,522.00 *gal*
Application event totals		43.26	5.52	58.24	570.60	

F1 - 04/10/2022: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/23/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	37.85	4.83	50.96	326.66	126,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	165.11	1,521,953.00 *gal*
Application event totals		37.85	4.83	50.96	491.77	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/30/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	159.10	1,466,550.00 *gal*
Application event totals		0.00	0.00	0.00	159.10	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/06/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	64.89	8.28	87.36	559.98	216,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	114.90	1,059,175.00 *gal*
Application event totals		64.89	8.28	87.36	674.89	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/14/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	43.26	5.52	58.24	373.32	144,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	123.74	1,140,650.00 *gal*
Application event totals		43.26	5.52	58.24	497.07	

F1 - 08/16/2022: Sudangrass, silage

Field name: F1

Crop: Sudangrass, silage

Plant date: 08/16/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

F1 - 08/16/2022: Sudangrass, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/17/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	304.06	2,802,740.00 *gal*
Application event totals		0.00	0.00	0.00	304.06	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/04/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	75.71	9.66	101.92	653.31	252,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	174.36	1,607,200.00 *gal*
Application event totals		75.71	9.66	101.92	827.67	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/16/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	48.67	6.21	65.52	419.99	162,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	322.09	2,968,949.00 *gal*
Application event totals		48.67	6.21	65.52	742.07	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/04/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	133.05	1,226,410.00 *gal*
Application event totals		0.00	0.00	0.00	133.05	

F2 - 10/26/2021: Oats, silage-soft dough

Field name: F2

Crop: Oats, silage-soft dough

Plant date: 10/26/2021

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

F2 - 10/26/2021: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/14/2021	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	69.79	2.79	39.79	154.37	144,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	280.06	4,474,704.00 *gal*
Application event totals		69.79	2.79	39.79	434.43	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
12/08/2021	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	52.34	2.09	29.85	115.78	108,000.00 *gal*
Application event totals		52.34	2.09	29.85	115.78	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/19/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	30.50	3.91	45.55	346.64	216,000.00 *gal*
Application event totals		30.50	3.91	45.55	346.64	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/28/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	12.71	1.63	18.98	144.43	90,000.00 *gal*
Application event totals		12.71	1.63	18.98	144.43	

F2 - 04/10/2022: Corn, silage

Field name: F2

Crop: Corn, silage

Plant date: 04/10/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

F2 - 04/10/2022: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/31/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	45.76	5.87	68.32	519.96	324,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	148.90	2,379,070.00 *gal*
Application event totals		45.76	5.87	68.32	668.86	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/07/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	40.56	5.17	54.60	349.99	234,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	238.24	3,806,512.00 *gal*
Application event totals		40.56	5.17	54.60	588.23	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/19/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	213.15	3,405,655.00 *gal*
Application event totals		0.00	0.00	0.00	213.15	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/27/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	169.30	2,704,970.00 *gal*
Application event totals		0.00	0.00	0.00	169.30	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/05/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	49.92	6.37	67.20	430.76	288,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	136.66	2,183,530.00 *gal*
Application event totals		49.92	6.37	67.20	567.42	

F2 - 04/10/2022: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/14/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	163.18	2,607,200.00 *gal*
Application event totals		0.00	0.00	0.00	163.18	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/23/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	152.98	2,444,250.00 *gal*
Application event totals		0.00	0.00	0.00	152.98	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/28/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	37.44	4.78	50.40	323.07	216,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	132.58	2,118,350.00 *gal*
Application event totals		37.44	4.78	50.40	455.65	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/09/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	74.87	9.55	100.80	646.13	432,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	132.58	2,118,350.00 *gal*
Application event totals		74.87	9.55	100.80	778.72	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/16/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	53.04	6.77	71.40	457.68	306,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	188.51	3,011,970.00 *gal*
Application event totals		53.04	6.77	71.40	646.19	

F2 - 08/16/2022: Sudangrass, silage

Field name: F2

Crop: Sudangrass, silage

Plant date: 08/16/2022

F2 - 08/16/2022: Sudangrass, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/17/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	226.00	3,610,972.00 *gal*
Application event totals		0.00	0.00	0.00	226.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/02/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	74.87	9.55	100.80	646.13	432,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	169.30	2,704,970.00 *gal*
Application event totals		74.87	9.55	100.80	815.43	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/17/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	62.39	7.96	84.00	538.45	360,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	197.85	3,161,230.00 *gal*
Application event totals		62.39	7.96	84.00	736.30	

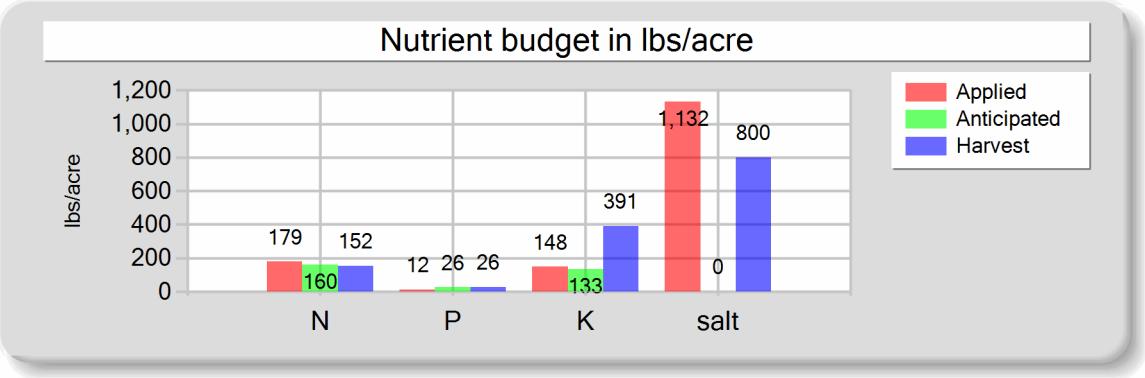
Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/20/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	147.24	2,352,485.00 *gal*
Application event totals		0.00	0.00	0.00	147.24	

B. NUTRIENT BUDGET

F1 - 10/26/2021: Oats, silage-soft dough

Field name: F1 Crop: Oats, silage-soft dough Plant date: 10/26/2021



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	173.84	11.62	147.93	868.41
Fresh water	0.00	0.00	0.00	263.82
Atmospheric deposition	4.67	0.00	0.00	0.00
Total nutrients applied	178.51	11.62	147.93	1,132.23
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	152.15	26.18	391.00	799.51
Nutrient balance	26.36	-14.56	-243.08	332.72
Applied to removed ratio	1.17	0.44	0.38	1.42

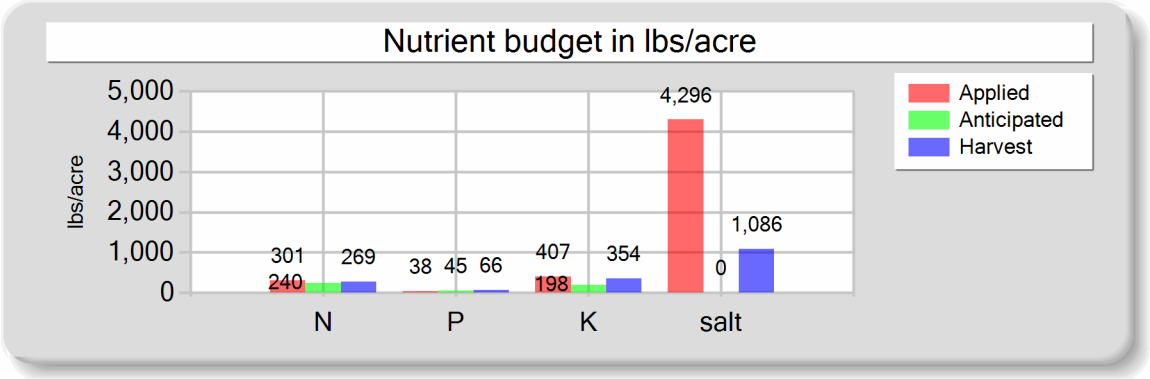
Fresh water applied	
2,431,815.00	*gallons*
89.56	*acre-inches*
5.97	*inches/acre*

Process wastewater applied	
360,000.00	*gallons*
13.26	*acre-inches*
0.88	*inches/acre*

Total harvests for the crop	
1	*harvests*

F1 - 04/10/2022: Corn, silage

Field name: F1 Crop: Corn, silage Plant date: 04/10/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	296.21	37.83	406.56	2,700.78
Fresh water	0.00	0.00	0.00	1,595.58
Atmospheric deposition	4.67	0.00	0.00	0.00
Total nutrients applied	300.88	37.83	406.56	4,296.36
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	268.71	65.54	353.92	1,085.58
Nutrient balance	32.17	-27.71	52.64	3,210.78
Applied to removed ratio	1.12	0.58	1.15	3.96

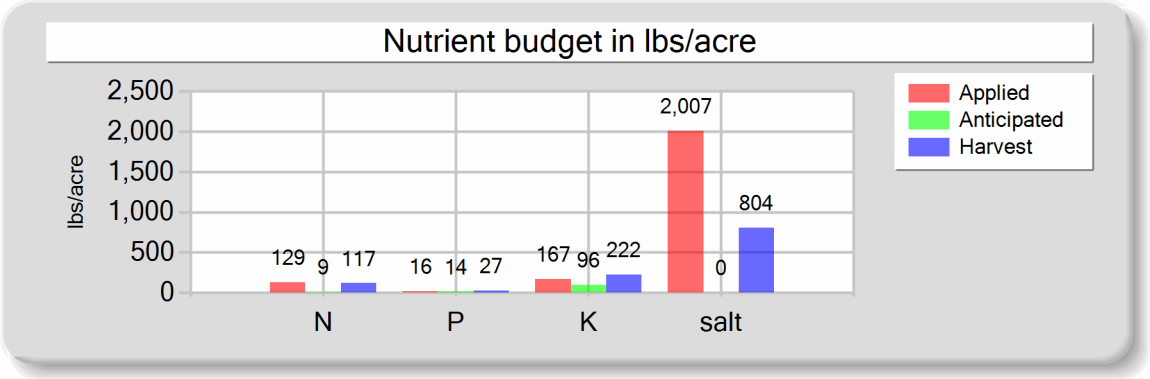
Fresh water applied	
14,707,867.00	*gallons*
541.64	*acre-inches*
36.11	*inches/acre*

Process wastewater applied	
1,026,000.00	*gallons*
37.78	*acre-inches*
2.52	*inches/acre*

Total harvests for the crop	
1	*harvests*

F1 - 08/16/2022: Sudangrass, silage

Field name: F1 Crop: Sudangrass, silage Plant date: 08/16/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	124.37	15.87	167.44	1,073.30
Fresh water	0.00	0.00	0.00	933.55
Atmospheric deposition	4.67	0.00	0.00	0.00
Total nutrients applied	129.04	15.87	167.44	2,006.85
Anticipated crop nutrient removal	8.80	13.60	96.00	0.00
Actual crop nutrient removal	116.62	26.79	222.22	804.36
Nutrient balance	12.42	-10.92	-54.77	1,202.49
Applied to removed ratio	1.11	0.59	0.75	2.49

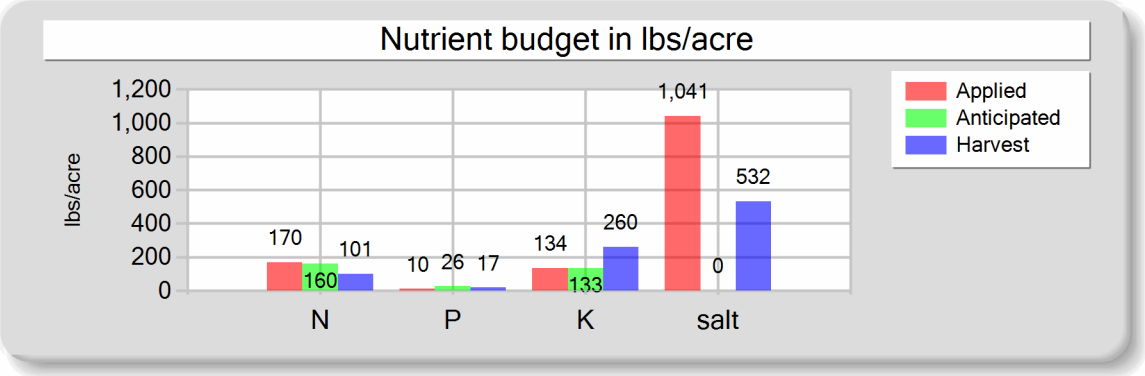
Fresh water applied	
8,605,299.00	*gallons*
316.90	*acre-inches*
21.13	*inches/acre*

Process wastewater applied	
414,000.00	*gallons*
15.25	*acre-inches*
1.02	*inches/acre*

Total harvests for the crop	
1	*harvests*

F2 - 10/26/2021: Oats, silage-soft dough

Field name: F2 Crop: Oats, silage-soft dough Plant date: 10/26/2021



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	165.35	10.42	134.17	761.22
Fresh water	0.00	0.00	0.00	280.06
Atmospheric deposition	4.67	0.00	0.00	0.00
Total nutrients applied	170.01	10.42	134.17	1,041.28
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	101.18	17.41	260.03	531.71
Nutrient balance	68.83	-6.98	-125.87	509.57
Applied to removed ratio	1.68	0.60	0.52	1.96

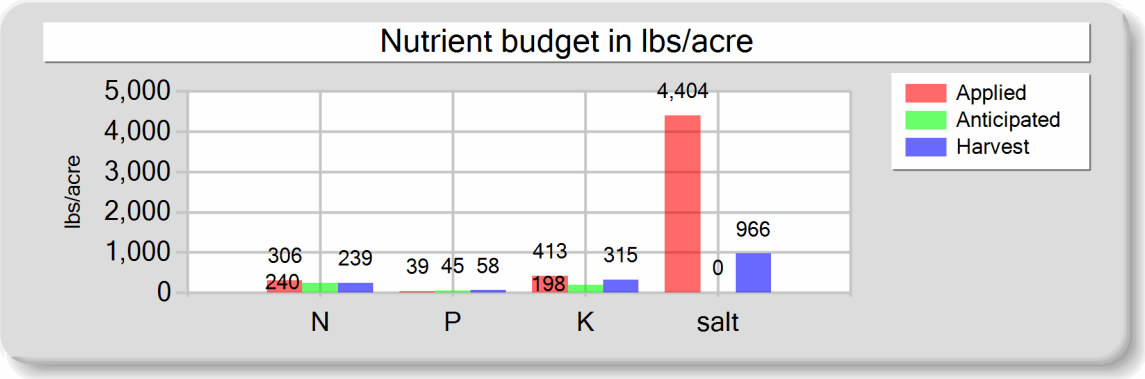
Fresh water applied	
4,474,704.00	*gallons*
164.79	*acre-inches*
6.34	*inches/acre*

Process wastewater applied	
558,000.00	*gallons*
20.55	*acre-inches*
0.79	*inches/acre*

Total harvests for the crop	
1	*harvests*

F2 - 04/10/2022: Corn, silage

Field name: F2 Crop: Corn, silage Plant date: 04/10/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	301.58	38.51	412.73	2,727.58
Fresh water	0.00	0.00	0.00	1,676.08
Atmospheric deposition	4.67	0.00	0.00	0.00
Total nutrients applied	306.24	38.51	412.73	4,403.67
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	239.19	58.34	315.04	966.32
Nutrient balance	67.05	-19.83	97.69	3,437.35
Applied to removed ratio	1.28	0.66	1.31	4.56

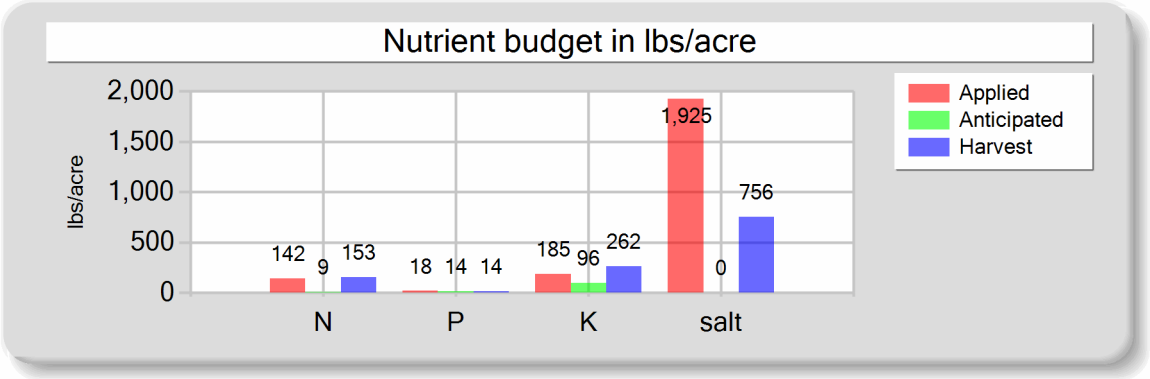
Fresh water applied	
26,779,857.00	*gallons*
986.21	*acre-inches*
37.93	*inches/acre*

Process wastewater applied	
1,800,000.00	*gallons*
66.29	*acre-inches*
2.55	*inches/acre*

Total harvests for the crop	
1	*harvests*

F2 - 08/16/2022: Sudangrass, silage

Field name: F2 Crop: Sudangrass, silage Plant date: 08/16/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	137.27	17.51	184.80	1,184.58
Fresh water	0.00	0.00	0.00	740.39
Atmospheric deposition	4.67	0.00	0.00	0.00
Total nutrients applied	141.94	17.51	184.80	1,924.97
Anticipated crop nutrient removal	8.80	13.60	96.00	0.00
Actual crop nutrient removal	152.87	14.18	261.62	755.63
Nutrient balance	-10.94	3.33	-76.81	1,169.34
Applied to removed ratio	0.93	1.23	0.71	2.55

Fresh water applied	
11,829,657.00	*gallons*
435.65	*acre-inches*
16.76	*inches/acre*

Process wastewater applied	
792,000.00	*gallons*
29.17	*acre-inches*
1.12	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

Fall 2021 Separated Solids

Sample and source description: Fall 2021 Separated Solids

Sample date: 11/16/2021 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 76.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	19,700.00	1,700.00	1,000.00	8,500.00	2,700.00	3,500.00	2,000.00	686.00		8.48
DL	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00		0.01

Spring Separated Solids

Sample and source description: Spring Separated Solids

Sample date: 04/25/2022 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 74.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	22,700.00	6,200.00	6,800.00							65.20
DL	100.00	100.00	100.00							0.01

Fall Separated Solids

Sample and source description: Fall Separated Solids

Sample date: 11/14/2022 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 78.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,200.00	1,940.00	4,770.00	8,680.00	2,870.00	1,260.00	3,000.00	5,680.00		10.10
DL	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00		0.01

B. PROCESS WASTEWATER ANALYSES

2021 WWSQ4

Sample and source description: 2021 WWSQ4

Sample date: 10/29/2021 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1,510.00	589.00	0.00	0.00	60.40	861.00								6,660.00	3,340
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

WWSQ1

Sample and source description: WWSQ1

Sample date: 03/29/2022 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	440.00	376.00	0.00	0.00	56.40	657.00								7,050.00	5,000
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

WWSQ2

Sample and source description: WWSQ2

Sample date: 06/21/2022 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1,370.00	305.00	0.00	0.00	56.90	931.00								7,500.00	3,840
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

WWSQ3

Sample and source description: WWSQ3

Sample date: 09/27/2022 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	540.00	297.00	0.00	0.00	68.90	727.00	143.00	133.00	285.00	2,740.00	0.00	256.00	325.00	5,540.00	4,660
DL	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	1.00	10

WWSQ4

Sample and source description: WWSQ4

Sample date: 12/12/2022 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	613.00	328.00	0.00	0.00	92.40	680.00								5,370.00	5,140
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

C. FRESH WATER ANALYSES

T.I.D Canal

Canal

Sample description: Canal

Sample date: 10/04/2022 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00								286.00	195
DL	0.50	0.50	0.50								1.00	10

D. SOIL ANALYSES

No soil analyses entered.

E. PLANT TISSUE ANALYSES

F1 - 10/26/2021: Oats, silage-soft dough

F1 - 10/26/2021: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 04/13/2022 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 63.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	9,300.00	1,600.00	23,900.00		13.50
DL	100.00	100.00	100.00		0.01

F1 - 04/10/2022: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 08/23/2022 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 67.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,100.00	1,000.00	5,400.00		5.16
DL	100.00	100.00	100.00		0.01

F1 - 08/16/2022: Sudangrass, silage

Sudan Silage

Sample and source description: Sudan Silage

Sample date: 11/08/2022 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 69.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	7,400.00	1,700.00	14,100.00		16.90
DL	100.00	100.00	100.00		0.01

F2 - 10/26/2021: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 04/13/2022 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 63.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	9,300.00	1,600.00	23,900.00		13.50
DL	100.00	100.00	100.00		0.01

F2 - 04/10/2022: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 08/23/2022 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 67.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,100.00	1,000.00	5,400.00		5.16
DL	100.00	100.00	100.00		0.01

F2 - 08/16/2022: Sudangrass, silage

Sudan Silage

Sample and source description: Sudan Silage

Sample date: 11/08/2022 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 60.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	9,700.00	900.00	16,600.00		12.20
DL	100.00	100.00	100.00		0.01

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

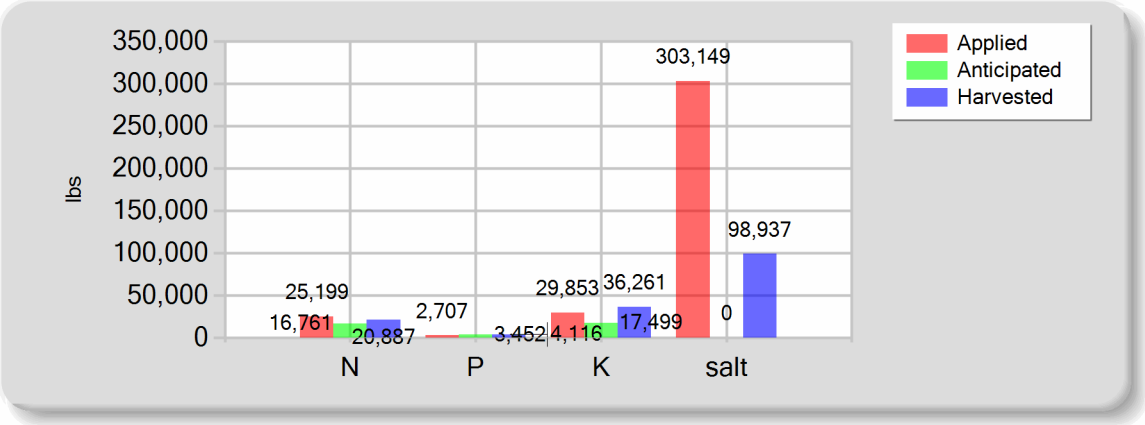
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

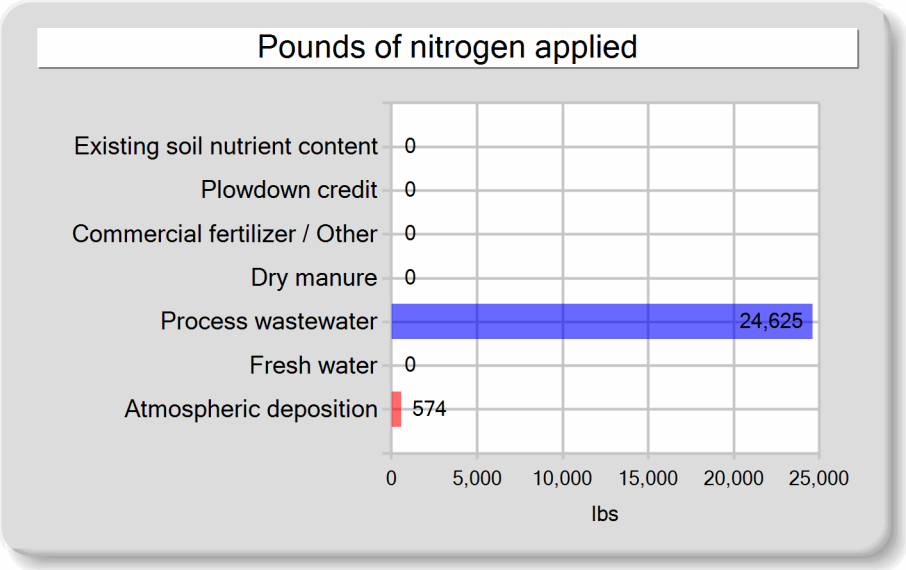
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

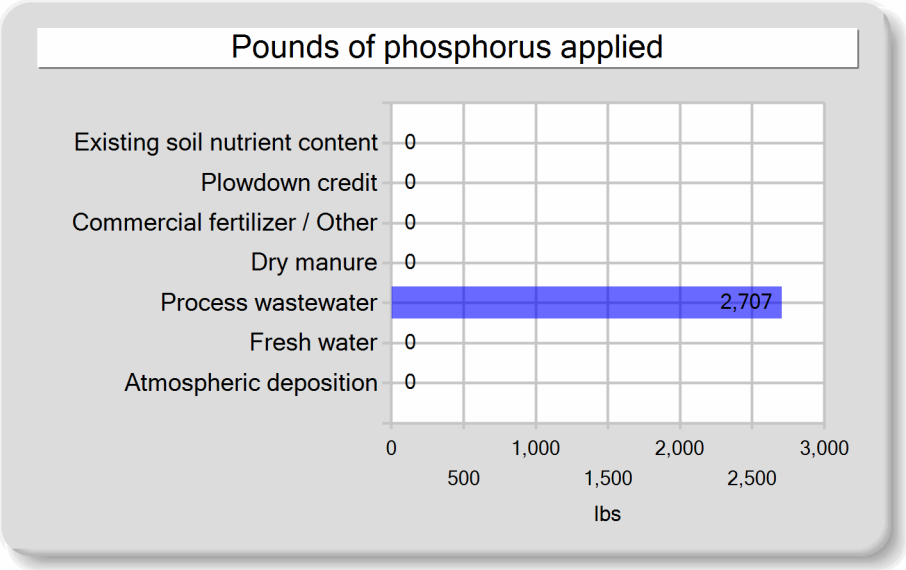
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	24,625.43	2,707.23	29,853.19	191,145.23
Fresh water	0.00	0.00	0.00	112,004.03
Atmospheric deposition	574.00	0.00	0.00	0.00
Total nutrients applied	25,199.43	2,707.23	29,853.19	303,149.26
Anticipated crop nutrient removal	16,760.80	3,452.20	17,498.80	0.00
Actual crop nutrient removal	20,886.79	4,115.85	36,260.82	98,936.79
Nutrient balance	4,312.64	-1,408.62	-6,407.64	204,212.48
Applied to removed ratio	1.21	0.66	0.82	3.06

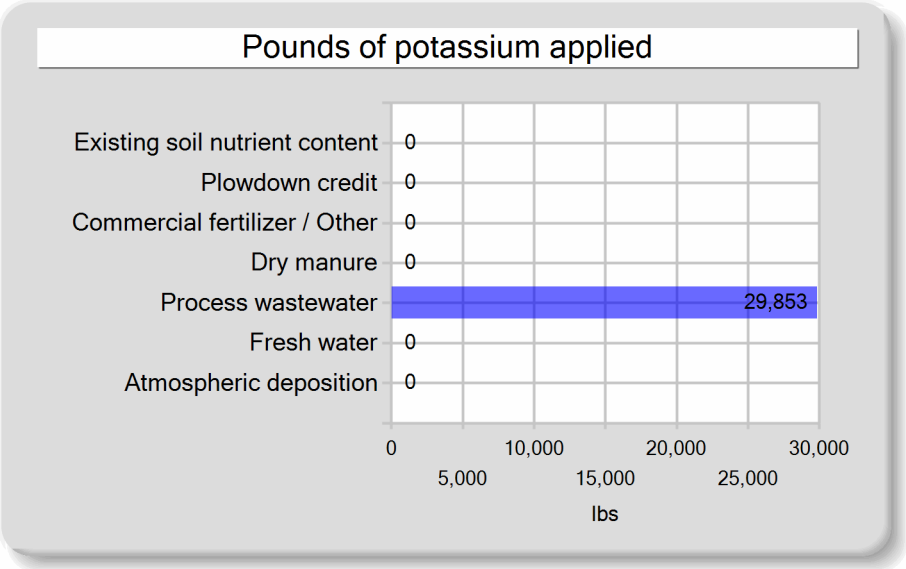
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

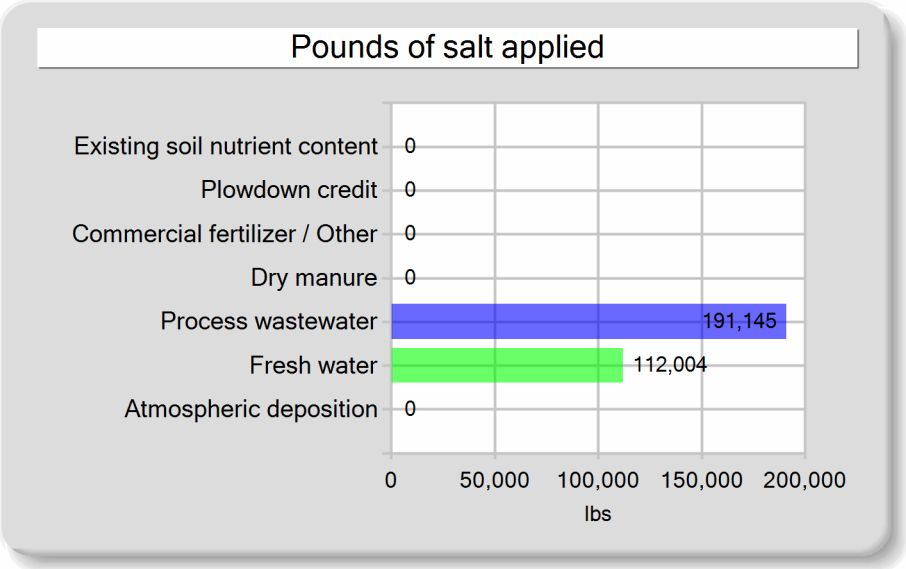


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	No
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

Extremely lower than anticipated yields for winter forage (16 tons per acre anticipated vs 5.4 tons per acre removed) caused Field 1 ratio to be above the 1.4 threshold. Applications were in line with targeted yields and nitrogen removal.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

Maia Cipponeri (May 22, 2024 14:41 PDT)	Maia Cipponeri (May 22, 2024 14:41 PDT)
SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
R & R Ranches LLC	Vierra Dairy LLC
PRINT OR TYPE NAME	PRINT OR TYPE NAME
May 22, 2024	May 22, 2024
DATE	DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

www.denelelabs.com

Compliance Analysis Report

Customer
Vierra Dairy Farms #2
P.O. Box 1350
Hilmar, CA 95324

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 1/30/24 10:10 AM
SUBMITTED BY: KK
ANALYZED DATE: 1/31/24

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 16.0c and passed hold time.

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
DW 1	T4030006A	Electrical Conductivity (EC)	EPA 120.1 and	0.953	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	0.235	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	610	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	103	101 (T4030004A)	98.8 (T4030004A)	1.80
Electrical Conductivity (EC)	99.1			0.959
Nitrate Nitrogen (NO3-N)	105	106 (T4030002A)	104 (T4030002A)	1.46

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

ELAP Certificate No.: 2714

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

www.denelelabs.com

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

Compliance Analysis Report

Customer
Vierra Dairy Farms #2
P.O. Box 1350
Hilmar, CA 95324

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 1/30/24 10:10 AM
SUBMITTED BY: KK
ANALYZED DATE: 1/31/24

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 16.6c and passed hold time.

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
DW 2	T4030006B	Electrical Conductivity (EC)	EPA 120.1 and	0.960	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	0.485	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	614	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	103	101 (T4030004A)	98.8 (T4030004A)	1.80
Electrical Conductivity (EC)	99.1			0.959
Nitrate Nitrogen (NO3-N)	105	106 (T4030002A)	104 (T4030002A)	1.46

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

ELAP Certificate No.: 2714

DENELE ANALYTICAL, INC.

865 S. KILROY
TURLOCK, CA 95380

PH# 209-634-9055 FAX# 209-634-9057

DAIRY COMPLIANCE

Client: VIERRA DAIRY FARMS #2
Mailing address:
(not facility address)
Phone#:

Authorized Copy Released To: F:R
Sampled By: JM
Project ID:

ANALYSIS TO BE COMPLETED:

IRRIGATION/GROUND WATER
- H1 EC*, NO3-N, NH4-N*
- H6 EC, TN, TDS
- H7 Combination of H1 & H6
- H14 EC, NO3-N, NH4-N, Ca, Mg, Na, Bicarb Carb, SO4, Cl, TDS

PROCESS WASTE WATER
- H3 EC*, NH4-N, TKN, TP, K, TDS
- H4 EC*, NO3-N, NH4-N, TKN, TP, K, TDS
- H5 Ca, Mg, Na, Bicarb, Carb, SO4, Cl
- H8 Combination of H3 &H5
- H9 Combination of H4 &H5

TILE DRAIN
- H2 EC*, NO3-N, TP, NH4-N*, TDS

** Required Field Measurement*

___ (Check here if performed)

DISCHARGE WATER
- E1 EC*, Temp*, pH*, NO3-N, NH4-N, TKN, TP, K, TDS, Dissolved Oxygen*, BOD 5, TSS, Total & Fecal Coliforms
- E2 EC,* Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Dissolved Oxygen*, Total & Fecal Coliforms
- E3 EC*, Temp*, pH*, NO3-N, NH4-N*, TP, Total & Fecal Coliforms
- E4 EC*, Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Total & Fecal Coliform

MANURE
- M1 % Moisture
- M3 TN TP, K, % Moisture
- M4 Ca, Mg, Na, Bicarb, Carb, SO4, Cl, Ash
- M7 Combination of M3 & M4

PLANT TISSUE
- F18 % Moisture, Ash, TN, P, K, DM and as received
- F18+ % Moisture, Ash, TN, P, K, DM and as received (Package also includes NIR DCAD)
- F19+ % Moisture, TN (Package includes complete plant tissue analysis)

SOIL
- K3 Soluble Phosphorus
- K4 1ft NO3, Organic Matter
- K5 Pre-Plant (2ft NO3), Fall (1-2ft NO3), Fall (2-3ft NO3)
- K6 0-1ft EC, NO3, Soluble P, K, Organic Matter
- K7 Combination of K3 & K4
- K8 Combination of K3 & K5

	Date Collected	Time Collected	Sample ID	Description	Temp	Well Purge	Analysis: H1	H3	M3	F18	OTHER
1	12/31	10:30	Domestic	DW1	16.0°c		X				
2	12/31	10:45	wells	DW2	16.6°c		X				
3											
4											
5											
6											
7											
8											
9											
10											

CHAIN OF CUSTODY RECORDING

	Signature	Company	Received Date & Time	Relinquished Date & Time	(LABORATORY USE ONLY) Logged in By:	LABORATORY NUMBER
1st	[illegible]	FRG				
2nd	[illegible]	DENELE ANALYTICAL, INC.	1-30-24 10:10			T403006

Required to be submitted to any agency? (YES / NO)

2023 Annual Report-Vierra Dairy Farms #2

Final Audit Report 2024-05-22

Created:	2024-05-22
By:	Kristen Jones (kjones@fragservices.com)
Status:	Signed
Transaction ID:	CBJCHBCAABAAaEbICb5tx_q7EtrfKfN2ABfFs5Vk9mvL

"2023 Annual Report-Vierra Dairy Farms #2" History

Document created by Kristen Jones (kjones@fragservices.com)
2024-05-22 - 2:59:51 PM GMT

Document emailed to Maia Cipponeri (maia@vierradairy.com) for signature
2024-05-22 - 3:00:30 PM GMT

Email viewed by Maia Cipponeri (maia@vierradairy.com)
2024-05-22 - 3:44:07 PM GMT

Document e-signed by Maia Cipponeri (maia@vierradairy.com)
Signature Date: 2024-05-22 - 9:41:40 PM GMT - Time Source: server

Agreement completed.
2024-05-22 - 9:41:40 PM GMT

Adobe Acrobat Sign